

Statistical Hypothesis Testing about Regression and Correlation Analysis

Lecture Notes

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Abstract

These lecture notes provide a comprehensive overview of statistical hypothesis testing with applications to regression analysis, correlation analysis, and variance comparisons. The material covers theoretical foundations, step-by-step procedures, practical examples, and exercises to reinforce learning. Emphasis is placed on understanding the logic of hypothesis testing, proper interpretation of results, and real-world applications.

1 Introduction to Hypothesis Testing

1.1 What is Hypothesis Testing?

Hypothesis testing is a systematic statistical procedure used to determine whether there is enough evidence in a sample of data to infer that a certain condition is true for the entire population. It provides a structured framework for making decisions based on data while controlling for the probability of errors.

1.2 The Six-Step Procedure

Hypothesis Testing Framework

Step 1: Formulate Hypotheses

- **Null Hypothesis (H_0)** : *The default assumption (e.g., "no effect", "no difference", "no relationship")*
- **Alternative Hypothesis (H_1)** : *What we aim to prove (e.g., "there is an effect", "there is a difference")*

Step 2: Set Level of Significance (α)

- Probability of rejecting H_0 when it is true (*Type I Error*)
- Common values: $\alpha = 0.05$ (5%) or $\alpha = 0.01$ (1%)

Step 3: Select Test Statistic

- A numerical value calculated from sample data
- Depends on the type of test (t-statistic, F-statistic, etc.)

Step 4: Define Critical Region

- Range of values for which we reject H_0
- Depends on α and the direction of H_1

Step 5: Perform Calculations

- Compute test statistic from sample data
- Calculate necessary parameters

Step 6: Draw Conclusion

- Compare test statistic to critical value
- Reject H_0 if test statistic falls in critical region

Statistical Concept	Courtroom Analogy
Null Hypothesis (H_0)	Defendant is innocent
Alternative Hypothesis (H_1)	Defendant is guilty
Sample Data	Evidence presented
Level of Significance (α)	Standard of proof ("beyond reasonable doubt")
Test Statistic	Strength of evidence
Critical Region	Threshold for conviction
Conclusion	Verdict (guilty/not guilty)

Table 1: Analogy between hypothesis testing and courtroom proceedings

1.3 Analogy: Courtroom Trial

2 Regression Analysis

2.1 Introduction to Regression

Regression analysis is a statistical method used to model and examine the relationship between a dependent variable (Y) and one or more independent variables (X). In simple linear regression, we find the straight line that best fits the data points.

2.2 The Regression Equation

The estimated regression line is expressed as:

$$\hat{Y} = a + bX$$

where:

- $\hat{Y}$ = predicted value of Y
- a = intercept (value of Y when $X = 0$)
- b = slope/regression coefficient (change in Y for a 1-unit change in X)

2.3 Key Calculations

2.3.1 Slope (b)

$$b = \frac{n \sum XY - \sum X \sum Y}{n \sum X^2 - (\sum X)^2}$$

2.3.2 Intercept (a)

$$a = \bar{Y} - b\bar{X}$$

where $\bar{Y} = \frac{\sum Y}{n}$ and $\bar{X} = \frac{\sum X}{n}$

2.3.3 Standard Error of Estimate (S_{yx})

Measures dispersion of data points around the regression line:

$$S_{yx} = \sqrt{\frac{\sum Y^2 - a \sum Y - b \sum XY}{n - 2}}$$

2.3.4 Standard Error of Slope (S_b)

Measures reliability of the slope estimate:

$$S_b = \frac{S_{yx}}{\sqrt{\frac{\sum(X-\bar{X})^2}{n}}}$$
$$\sqrt{\frac{\sum(X-\bar{X})^2}{n}} = \sqrt{\frac{\sum X^2 - \frac{(\sum X)^2}{n}}{n}}$$

2.4 Example: Height vs. Weight Regression

Example Problem

The following data shows the height (cm) and weight (kg) of 5 individuals:

Height (X)	Weight (Y)	XY	X ²	Y ²
160	55	8800	25600	3025
165	60	9900	27225	3600
170	65	11050	28900	4225
175	70	12250	30625	4900
180	75	13500	32400	5625

Find: (a) Regression equation, (b) Standard error of estimate, (c) Test significance of slope at $\alpha = 0.05$

2.4.1 Step-by-Step Solution

Step 1: Calculate sums

$$n = 5, \quad \sum X = 850, \quad \sum Y = 325, \quad \sum XY = 55500, \quad \sum X^2 = 144750, \quad \sum Y^2 = 21375$$

Step 2: Calculate slope (b)

$$b = \frac{5(55500) - (850)(325)}{5(144750) - (850)^2} = \frac{277500 - 276250}{723750 - 722500} = \frac{1250}{1250} = 1$$

Step 3: Calculate means

$$\bar{X} = \frac{850}{5} = 170, \quad \bar{Y} = \frac{325}{5} = 65$$

Step 4: Calculate intercept (a)

$$a = 65 - 1(170) = -105$$

Step 5: Regression equation

$$\hat{Y} = -105 + 1X$$

Step 6: Standard error of estimate

$$S_{yx} = \sqrt{\frac{21375 - (-105)(325) - 1(55500)}{5 - 2}} = \sqrt{\frac{21375 + 34125 - 55500}{3}} = \sqrt{\frac{0}{3}} = 0$$

Step 7: Test hypothesis $H_0: \beta = 0$ vs $H_1: \beta \neq 0$

$$t = \frac{b - 0}{S_b} = \frac{1}{0} \quad (\text{undefined in this perfect fit case})$$

2.5 Hypothesis Testing for Regression Coefficient

Hypotheses:

- $H_0 : \beta = 0$ (no relationship between X and Y)
- $H_1 : \beta \neq 0$ (significant relationship)

Test statistic:

$$t = \frac{b - 0}{S_b}$$

Degrees of freedom: $df = n - 2$

Decision rule: Reject H_0 if $|t| > t_{\alpha/2, n-2}$

3 Example 21: Hypothesis Testing for Regression Coefficient

3.1 Problem Statement

We want to test the hypothesis about the population regression coefficient β . The null hypothesis is that β equals a specific value β_0 , and the alternative hypothesis is that β is not equal to β_0 .

3.2 Hypotheses

$$H_0 : \beta = \beta_0 \quad \text{vs} \quad H_1 : \beta \neq \beta_0$$

3.3 Testing Procedure

1. **Level of Significance:** $\alpha = 0.05$ or as given

2. **Test Statistic:**

$$t = \frac{b - \beta_0}{s_b}$$

where:

- b = sample regression coefficient
- β_0 = hypothesized population regression coefficient
- s_b = standard error of b

3. **Critical Region:**

- For $H_1 : \beta \neq \beta_0$: $|t| > t_{\alpha/2, (n-2)}$
- For $H_1 : \beta > \beta_0$: $t > t_{\alpha, (n-2)}$
- For $H_1 : \beta < \beta_0$: $t < -t_{\alpha, (n-2)}$

4. **Calculations:**

Step 1: Calculate b (sample regression coefficient):

$$b = \frac{n \sum xy - \sum x \sum y}{n \sum x^2 - (\sum x)^2}$$

Step 2: Calculate S_{xy} (standard error of estimate):

$$S_{xy} = \sqrt{\frac{\sum y^2 - a \sum y - b \sum xy}{n - 2}}$$

where a is the intercept: $a = \bar{y} - b\bar{x}$

Step 3: Calculate standard deviation of x :

$$\sqrt{\frac{\sum (x - \bar{x})^2}{n}} = \sqrt{\frac{\sum x^2 - (\sum x)^2}{n}}$$

Step 4: Calculate S_b (standard error of b):

$$S_b = \frac{S_{xy}}{\sqrt{\frac{\sum (x - \bar{x})^2}{n}}}$$

Step 5: Calculate test statistic:

$$t = \frac{b - \beta_0}{S_b}$$

5. **Conclusion:** Reject H_0 if the calculated t -value falls in the rejection region; otherwise, do not reject H_0 .

4 Example 22: Regression Line Calculation and Hypothesis Test

4.1 Problem Statement

Given the following summary statistics from a sample of size $n = 12$:

$$\sum x = 766, \quad \sum y = 1700, \quad \sum xy = 109380, \quad \sum x^2 = 49068, \quad \sum y^2 = 246100$$

We need to:

1. Calculate the regression line
2. Test the hypothesis $H_0 : \beta = 0$ vs $H_1 : \beta \neq 0$ at $\alpha = 0.05$

4.2 Step-by-Step Solution

Step 1: Calculate Regression Coefficient b

$$b = \frac{n \sum xy - \sum x \sum y}{n \sum x^2 - (\sum x)^2} = \frac{(12 \times 109380) - (766 \times 1700)}{(12 \times 49068) - (766)^2}$$

$$b = \frac{1,312,560 - 1,302,200}{588,816 - 586,756} = \frac{10,360}{2,060} = 5.029$$

Step 2: Calculate Means

$$\bar{x} = \frac{\sum x}{n} = \frac{766}{12} = 63.83$$

$$\bar{y} = \frac{\sum y}{n} = \frac{1700}{12} = 141.667$$

Step 3: Calculate Intercept a

$$a = \bar{y} - b\bar{x} = 141.667 - (5.029 \times 63.83)$$

$$a = 141.667 - 321.025 = -179.358$$

Step 4: Write Regression Equation

The estimated regression line is:

$$\hat{y} = -179.358 + 5.029x$$

Step 5: Calculate Standard Error of Estimate S_{yx}

$$S_{yx} = \sqrt{\frac{\sum y^2 - a \sum y - b \sum xy}{n - 2}}$$

$$S_{yx} = \sqrt{\frac{246100 - (-179.358 \times 1700) - (5.029 \times 109380)}{10}}$$

$$S_{yx} = \sqrt{\frac{246100 + 304,908.6 - 550,054.02}{10}}$$

$$S_{yx} = \sqrt{\frac{954.58}{10}} = \sqrt{95.458} = 9.6169$$

Step 6: Calculate Standard Deviation of x

$$\sqrt{\sum (x - \bar{x})^2} = \sqrt{\frac{\sum x^2 - (\sum x)^2}{n}}$$

$$= \sqrt{\frac{49068 - \frac{(766)^2}{12}}{12}} = \sqrt{\frac{49068 - \frac{586,756}{12}}{12}}$$

$$= \sqrt{\frac{49068 - 48,896.33}{12}} = \sqrt{171.67} = 13.102$$

Step 7: Calculate Standard Error of b (S_b)

$$S_b = \frac{S_{yx}}{\sqrt{\sum (x - \bar{x})^2}} = \frac{9.6169}{13.102} = 0.7334$$

Step 8: Calculate Test Statistic

$$t = \frac{b - 0}{S_b} = \frac{5.029}{0.7334} = 6.852$$

Step 9: Determine Critical Value

For $\alpha = 0.05$, $df = n - 2 = 10$:

$$t_{0.025,10} = 2.228 \quad (\text{from t-table})$$

Step 10: Make Decision and Conclusion

Since $|t| = 6.852 > 2.228$, the calculated value falls in the rejection region.

Conclusion: We reject $H_0 : \beta = 0$. There is sufficient evidence to conclude that the population regression coefficient is significantly different from zero. This means there is a statistically significant linear relationship between x and y .

5 Interpretation and Applications

5.1 Interpretation of Results

- **Regression Coefficient** ($b = 5.029$): For every one-unit increase in x , y increases by approximately 5.029 units.
- **Intercept** ($a = -179.358$): When $x = 0$, the predicted value of y is -179.358. Note that this may or may not have practical meaning depending on the context.
- **Standard Error of Estimate** ($S_{yx} = 9.6169$): The typical distance between the observed y values and the predicted $\hat{y}$ values is about 9.617 units.
- **Test Statistic** ($t = 6.852$): The sample regression coefficient is 6.852 standard errors away from zero, indicating strong evidence against H_0 .

5.2 Practical Applications

This type of analysis can be applied in various fields:

Field	Application Example
Economics	Relationship between advertising expenditure and sales revenue
Medicine	Relationship between drug dosage and patient response
Education	Relationship between study hours and exam scores
Engineering	Relationship between temperature and material strength
Finance	Relationship between interest rates and stock prices

6 Correlation Analysis

6.1 Correlation Coefficient (r)

The correlation coefficient measures the strength and direction of linear relationship between two variables:

- Range: $-1 \leq r \leq 1$
- $r = 1$: perfect positive correlation
- $r = -1$: perfect negative correlation
- $r = 0$: no linear correlation

6.2 Calculating Correlation Coefficient

$$r = \frac{n \sum XY - \sum X \sum Y}{\sqrt{[n \sum X^2 - (\sum X)^2][n \sum Y^2 - (\sum Y)^2]}}$$

6.3 Hypothesis Testing for Correlation

Hypotheses:

- $H_0 : \rho = 0$ (no correlation in population)
- $H_1 : \rho \neq 0$ (significant correlation)

Test statistic:

$$t = \frac{r\sqrt{n-2}}{\sqrt{1-r^2}}$$

Degrees of freedom: $df = n - 2$

Decision rule: Reject H_0 if $|t| > t_{\alpha/2, n-2}$

6.4 Example: Study Hours vs. Exam Scores

Example Problem

A researcher collects data from 8 students on hours spent on social media (X) and GPA (Y):

Student	Hours (X)	GPA (Y)	XY	X ²	Y ²
1	2	3.8	7.6	4	14.44
2	3	3.6	10.8	9	12.96
3	4	3.5	14.0	16	12.25
4	5	3.2	16.0	25	10.24
5	6	3.0	18.0	36	9.00
6	7	2.8	19.6	49	7.84
7	8	2.5	20.0	64	6.25
8	9	2.0	18.0	81	4.00

Test if the correlation is significant at $\alpha = 0.01$

6.4.1 Step-by-Step Solution

Step 1: Calculate sums

$$n = 8, \quad \sum X = 44, \quad \sum Y = 24.4, \quad \sum XY = 124, \quad \sum X^2 = 284, \quad \sum Y^2 = 76.98$$

Step 2: Calculate correlation coefficient

$$r = \frac{8(124) - (44)(24.4)}{\sqrt{[8(284) - 44^2][8(76.98) - 24.4^2]}} = \frac{992 - 1073.6}{\sqrt{[2272 - 1936][615.84 - 595.36]}}$$
$$r = \frac{-81.6}{\sqrt{336 \times 20.48}} = \frac{-81.6}{\sqrt{6876.48}} = \frac{-81.6}{82.92} = -0.984$$

Step 3: Test significance ($\alpha = 0.01$)

$$t = \frac{-0.984\sqrt{8-2}}{\sqrt{1-(-0.984)^2}} = \frac{-0.984\sqrt{6}}{\sqrt{1-0.968}} = \frac{-2.410}{\sqrt{0.032}} = \frac{-2.410}{0.179} = -13.46$$

Step 4: Critical value

$$t_{0.005,6} = 3.707 \quad (\text{from t-table})$$

Step 5: Conclusion Since $|t| = 13.46 > 3.707$, we reject H_0 . There is a significant negative correlation between social media hours and GPA.

7 Example 23: Testing Significance of Population Correlation Coefficient

7.1 Problem Statement

A study examines the prices of a six-pack of cola (X) and a pound of chicken (Y) in eight different cities. Using $\alpha = 0.05$, determine whether the population correlation coefficient is significant.

7.2 Data

City	Hong Kong	London	Mumbai	Mexico	New York	Paris	Sydney
Cola Price (X)	3.27	2.22	2.28	3.04	2.33	2.69	4.07
Chicken Price (Y)	3.06	2.31	2.27	1.51	1.87	1.65	3.91

Table 2: Prices of cola and chicken in different cities

7.3 Summary Statistics

From the data, we have:

$$n = 8, \quad \sum X = 22.68, \quad \sum Y = 18.15, \quad \sum XY = 52.9, \quad \sum X^2 = 67.02, \quad \sum Y^2 = 43.61$$

7.4 Hypotheses

$$H_0 : \rho = 0 \quad (\text{No correlation in population})$$

$$H_1 : \rho \neq 0 \quad (\text{Significant correlation exists})$$

7.5 Level of Significance

$$\alpha = 0.05$$

7.6 Step-by-Step Solution

Step 1: Calculate Sample Correlation Coefficient (r)

$$\begin{aligned}r &= \frac{n \sum XY - \sum X \sum Y}{\sqrt{[n \sum X^2 - (\sum X)^2][n \sum Y^2 - (\sum Y)^2]}} \\r &= \frac{8(52.9) - (22.68)(18.15)}{\sqrt{[8(67.02) - (22.68)^2][8(43.61) - (18.15)^2]}} \\r &= \frac{423.2 - 411.642}{\sqrt{[536.16 - 514.3824][348.88 - 329.4225]}} \\r &= \frac{11.558}{\sqrt{[21.7776][19.4575]}} = \frac{11.558}{\sqrt{423.71}} = \frac{11.558}{20.584} = 0.561\end{aligned}$$

Step 2: Calculate Test Statistic (t)

$$\begin{aligned}t &= \frac{r\sqrt{n-2}}{\sqrt{1-r^2}} = \frac{0.561\sqrt{8-2}}{\sqrt{1-(0.561)^2}} \\t &= \frac{0.561\sqrt{6}}{\sqrt{1-0.3147}} = \frac{0.561 \times 2.449}{\sqrt{0.6853}} \\t &= \frac{1.374}{0.828} = 1.659\end{aligned}$$

Step 3: Determine Critical Value

For $\alpha = 0.05$ (two-tailed test) with $df = n - 2 = 6$:

$$t_{0.025,6} = 2.447 \quad (\text{from t-distribution table})$$

Step 4: Decision and Conclusion

Since $|t| = 1.659 < 2.447$, the calculated value falls in the acceptance region.

Conclusion: We do not reject H_0 . There is insufficient evidence to conclude that the population correlation coefficient is significantly different from zero. The correlation between cola and chicken prices is not statistically significant at the 5% level.

8 Example 24: Testing Regression and Correlation Coefficients

8.1 Problem Statement

In a linear regression problem, the following sums were computed from a random sample of size 10:

$$\sum x = 320, \quad \sum y = 250, \quad \sum x^2 = 12400, \quad \sum xy = 9415, \quad \sum y^2 = 7230$$

Test the following hypotheses:

- Using 5% level of significance, test that the population regression coefficient β_1 is greater than 0.

- ii) Test the hypothesis at 10% level of significance that the population regression coefficient is zero.

8.2 Part i: Test that $\beta_1 > 0$ at $\alpha = 0.05$

Hypotheses

$$H_0 : \beta = 0 \quad \text{vs} \quad H_1 : \beta > 0$$

One-tailed test (upper-tailed)

Step 1: Calculate Sample Regression Coefficient (b)

$$b = \frac{n \sum xy - \sum x \sum y}{n \sum x^2 - (\sum x)^2} = \frac{10(9415) - (320)(250)}{10(12400) - (320)^2}$$

$$b = \frac{94150 - 80000}{124000 - 102400} = \frac{14150}{21600} = 0.6551$$

Step 2: Calculate Intercept (a)

First calculate means:

$$\bar{x} = \frac{\sum x}{n} = \frac{320}{10} = 32, \quad \bar{y} = \frac{\sum y}{n} = \frac{250}{10} = 25$$

$$a = \bar{y} - b\bar{x} = 25 - 0.6551(32) = 25 - 20.9632 = 4.0368$$

Step 3: Calculate Standard Error of Estimate (S_{yx})

$$S_{yx} = \sqrt{\frac{\sum y^2 - a \sum y - b \sum xy}{n - 2}} = \sqrt{\frac{7230 - 4.0368(250) - 0.6551(9415)}{8}}$$

$$S_{yx} = \sqrt{\frac{7230 - 1009.2 - 6165.85}{8}} = \sqrt{\frac{54.95}{8}} = \sqrt{6.86875} = 2.621$$

Step 4: Calculate Standard Error of b (S_b)

$$S_b = \frac{S_{yx}}{\sqrt{\frac{\sum x^2 - (\sum x)^2/n}{n}}} = \frac{2.621}{\sqrt{\frac{12400 - (320)^2/10}{10}}}$$

$$S_b = \frac{2.621}{\sqrt{\frac{12400 - 10240}{10}}} = \frac{2.621}{\sqrt{216}} = \frac{2.621}{14.697} = 0.178$$

Step 5: Calculate Test Statistic (t)

$$t = \frac{b - 0}{S_b} = \frac{0.6551}{0.178} = 3.68$$

Step 6: Determine Critical Value

For $\alpha = 0.05$ (one-tailed) with $df = n - 2 = 8$:

$$t_{0.05,8} = 1.860 \quad (\text{from t-distribution table})$$

Step 7: Decision and Conclusion

Since $t = 3.68 > 1.860$, we reject H_0 .

Conclusion: There is sufficient evidence at the 5% significance level to conclude that the population regression coefficient is greater than zero.

8.3 Part ii: Test that $\beta = 0$ at $\alpha = 0.10$

Hypotheses

$$H_0 : \beta = 0 \quad \text{vs} \quad H_1 : \beta \neq 0$$

Two-tailed test

Step 1: Already have b and S_b from Part i

$$b = 0.6551, \quad S_b = 0.178$$

Step 2: Calculate Test Statistic (t)

$$t = \frac{b - 0}{S_b} = \frac{0.6551}{0.178} = 3.68$$

Step 3: Determine Critical Value

For $\alpha = 0.10$ (two-tailed) with $df = 8$:

$$t_{0.05,8} = 1.860 \quad (\text{from t-distribution table})$$

Step 4: Decision and Conclusion

Since $|t| = 3.68 > 1.860$, we reject H_0 .

Conclusion: There is sufficient evidence at the 10% significance level to conclude that the population regression coefficient is not equal to zero.

8.4 Additional Test: Correlation Coefficient at $\alpha = 0.05$

Hypotheses

$$H_0 : \rho = 0 \quad \text{vs} \quad H_1 : \rho \neq 0$$

Two-tailed test

Step 1: Calculate Sample Correlation Coefficient (r)

$$\begin{aligned} r &= \frac{n \sum xy - \sum x \sum y}{\sqrt{[n \sum x^2 - (\sum x)^2][n \sum y^2 - (\sum y)^2]}} \\ r &= \frac{10(9415) - (320)(250)}{\sqrt{[10(12400) - (320)^2][10(7230) - (250)^2]}} \\ r &= \frac{94150 - 80000}{\sqrt{[124000 - 102400][72300 - 62500]}} \\ r &= \frac{14150}{\sqrt{21600 \times 9800}} = \frac{14150}{\sqrt{211680000}} = \frac{14150}{14549.2} = 0.9726 \end{aligned}$$

Step 2: Calculate Test Statistic (t)

$$\begin{aligned} t &= \frac{r\sqrt{n-2}}{\sqrt{1-r^2}} = \frac{0.9726\sqrt{8}}{\sqrt{1-(0.9726)^2}} \\ t &= \frac{0.9726 \times 2.828}{\sqrt{1-0.9459}} = \frac{2.751}{\sqrt{0.0541}} = \frac{2.751}{0.233} = 11.81 \end{aligned}$$

Step 3: Determine Critical Value

For $\alpha = 0.05$ (two-tailed) with $df = 8$:

$$t_{0.025,8} = 2.306$$

Step 4: Decision and Conclusion

Since $|t| = 11.81 > 2.306$, we reject H_0 .

Conclusion: There is a statistically significant correlation between the variables at the 5% significance level.

9 Summary of Key Formulas

9.1 For Correlation Testing

- **Sample Correlation Coefficient:**

$$r = \frac{n \sum XY - \sum X \sum Y}{\sqrt{[n \sum X^2 - (\sum X)^2][n \sum Y^2 - (\sum Y)^2]}}$$

- **Test Statistic:**

$$t = \frac{r\sqrt{n-2}}{\sqrt{1-r^2}}$$

- **Degrees of Freedom:** $df = n - 2$

9.2 For Regression Testing

- **Regression Coefficient:**

$$b = \frac{n \sum xy - \sum x \sum y}{n \sum x^2 - (\sum x)^2}$$

- **Standard Error of Estimate:**

$$S_{yx} = \sqrt{\frac{\sum y^2 - a \sum y - b \sum xy}{n - 2}}$$

- **Standard Error of b:**

$$S_b = \frac{S_{yx}}{\sqrt{\sum x^2 - (\sum x)^2/n}}$$

- **Test Statistic:**

$$t = \frac{b - \beta_0}{S_b}$$

10 Practice Problems

10.1 Problem 1

Given: $n = 12$, $\sum X = 180$, $\sum Y = 240$, $\sum XY = 4000$, $\sum X^2 = 3000$, $\sum Y^2 = 5200$

Calculate:

- Correlation coefficient and test its significance at $\alpha = 0.05$
- Regression equation and test if the slope is significantly different from zero at $\alpha = 0.01$

10.2 Problem 2

A study examines the relationship between temperature (X in °C) and ice cream sales (Y in \$100s):

X	20	22	24	26	28	30	32
Y	15	18	22	25	30	35	40

Calculate:

- Correlation coefficient and test at $\alpha = 0.05$
- Regression equation
- Test $H_0 : \beta = 1$ vs $H_1 : \beta > 1$ at $\alpha = 0.05$

10.3 Problem 3

Given the regression equation $\hat{y} = 2.5 + 1.8x$ with $S_b = 0.3$ and $n = 15$:

Test $H_0 : \beta = 1.5$ vs $H_1 : \beta \neq 1.5$ at $\alpha = 0.05$

Answer Key (Brief Solutions)

Problem 1:

- $r = 0.923$, $t = 7.12 > t_{0.025,10} = 2.228$, reject H_0
- $b = 1.333$, $t = 8.0 > t_{0.005,10} = 3.169$, reject H_0

Problem 2:

(a) $r = 0.995, t = 19.9 > t_{0.025,5} = 2.571$, reject H_0

(b) $\hat{y} = -30.5 + 2.107x$

(c) $t = 3.69 > t_{0.05,5} = 2.015$, reject H_0

Problem 3:

$$t = \frac{1.8 - 1.5}{0.3} = 1.0 < t_{0.025,13} = 2.160, \text{ do not reject } H_0$$

10.4 Analogy: Kites Tethered Together

- **r 1:** Short, strong tether (kites move together)
- **r 0:** No tether (kites move independently)
- **r -1:** Opposite-direction tether (one goes up when other goes down)
- **Hypothesis testing:** Checking if the tether actually exists or if kites just happen to drift similarly

11 F-Test for Comparing Variances**11.1 When to Use F-Test**

The F-test compares variances of two populations to determine if they are significantly different. Applications include:

- Checking assumption of equal variances for t-tests
- Comparing consistency/precision between two processes
- Analyzing variability in different groups

11.2 Hypotheses

- $H_0 : \sigma_1^2 = \sigma_2^2$ (variances are equal)
- $H_1 : \sigma_1^2 \neq \sigma_2^2$ or $\sigma_1^2 > \sigma_2^2$ or $\sigma_1^2 < \sigma_2^2$

11.3 Test Statistic

$$F = \frac{S_1^2}{S_2^2} \quad (\text{always put larger variance in numerator})$$

where $S^2 = \frac{1}{n-1} \left[\sum X^2 - \frac{(\sum X)^2}{n} \right]$

11.4 Degrees of Freedom

- Numerator: $df_1 = n_1 - 1$
- Denominator: $df_2 = n_2 - 1$

11.5 Why Larger Variance in Numerator?

1. Ensures $F \geq 1$, making interpretation easier
2. Standard F-tables are constructed for $F \geq 1$
3. Simplifies decision: Reject H_0 if $F > \text{critical value}$

11.6 Example: Comparing Manufacturing Processes

Example Problem

Two manufacturing processes are tested for consistency:

Process A: 10, 12, 14, 15, 16 ($n_1 = 5$)

Process B: 8, 9, 10, 11, 12, 13, 14 ($n_2 = 7$)

Test if the variances are significantly different at $\alpha = 0.05$

11.6.1 Step-by-Step Solution

Step 1: Calculate variances

Process A:

$$\sum X_A = 67, \quad \sum X_A^2 = 921$$
$$S_A^2 = \frac{1}{5-1} \left[921 - \frac{(67)^2}{5} \right] = \frac{1}{4} \left[921 - \frac{4489}{5} \right] = \frac{1}{4} [921 - 897.8] = \frac{23.2}{4} = 5.8$$

Process B:

$$\sum X_B = 77, \quad \sum X_B^2 = 875$$
$$S_B^2 = \frac{1}{7-1} \left[875 - \frac{(77)^2}{7} \right] = \frac{1}{6} \left[875 - \frac{5929}{7} \right] = \frac{1}{6} [875 - 847] = \frac{28}{6} = 4.67$$

Step 2: F-statistic (larger variance in numerator)

$$F = \frac{5.8}{4.67} = 1.242$$

Step 3: Degrees of freedom

$$df_1 = 5 - 1 = 4, \quad df_2 = 7 - 1 = 6$$

Step 4: Critical value ($\alpha = 0.05$)

$$F_{0.05}(4, 6) = 4.53 \quad (\text{from F-table})$$

Step 5: Conclusion Since $1.242 < 4.53$, we do not reject H_0 . The variances are not significantly different.

12 Introduction to F-Test for Variances

The F-test is used to compare the variances of two populations. It is based on the ratio of two sample variances and follows the F-distribution under the null hypothesis. This test is commonly used to check the assumption of equal variances (homoscedasticity) before conducting tests like the two-sample t-test.

12.1 Hypotheses

- **Two-tailed test:** $H_0 : \sigma_1^2 = \sigma_2^2$ vs $H_1 : \sigma_1^2 \neq \sigma_2^2$
- **One-tailed test (upper):** $H_0 : \sigma_1^2 = \sigma_2^2$ vs $H_1 : \sigma_1^2 > \sigma_2^2$
- **One-tailed test (lower):** $H_0 : \sigma_1^2 = \sigma_2^2$ vs $H_1 : \sigma_1^2 < \sigma_2^2$

12.2 Test Statistic

$$F = \frac{S_1^2}{S_2^2}$$

where S_1^2 and S_2^2 are the sample variances. By convention, we place the larger sample variance in the numerator so that $F \geq 1$. This simplifies the comparison with critical values from the F-distribution table.

12.3 Degrees of Freedom

The F-distribution has two degrees of freedom:

- $df_1 = n_1 - 1$ (numerator)
- $df_2 = n_2 - 1$ (denominator)

12.4 Critical Region

- For a two-tailed test at level α , we compare F with $F_{\alpha/2}(df_1, df_2)$.
- For a one-tailed test, we compare F with $F_{\alpha}(df_1, df_2)$.

13 Example 1: Comparing Variances of Two Samples

13.1 Problem Statement

Two random samples are drawn from two normal populations:

Sample I: 20, 16, 36, 37, 33, 33, 18, 34, 35, 19

Sample II: 87, 33, 43, 35, 33, 34, 38, 88, 41, 43, 30, 37

Obtain the estimates of the population variances and test whether the two populations have the same variances at $\alpha = 0.05$.

13.2 Step-by-Step Solution

Step 1: State the Hypotheses

$$H_0 : \sigma_1^2 = \sigma_2^2 \quad \text{vs} \quad H_1 : \sigma_1^2 \neq \sigma_2^2$$

Two-tailed test.

Step 2: Organize the Data

We calculate the necessary sums:

For Sample I:

$$n_1 = 10, \quad \sum x_1 = 281, \quad \sum x_1^2 = 8469$$

For Sample II:

$$n_2 = 12, \quad \sum x_2 = 503, \quad \sum x_2^2 = 22599$$

Step 3: Calculate Sample Variances

$$\begin{aligned} S_1^2 &= \frac{1}{n_1 - 1} \left(\sum x_1^2 - \frac{(\sum x_1)^2}{n_1} \right) = \frac{1}{9} \left(8469 - \frac{(281)^2}{10} \right) \\ &= \frac{1}{9} \left(8469 - \frac{78961}{10} \right) = \frac{1}{9} (8469 - 7896.1) = \frac{1}{9} \times 572.9 = 63.66 \end{aligned}$$

$$\begin{aligned} S_2^2 &= \frac{1}{n_2 - 1} \left(\sum x_2^2 - \frac{(\sum x_2)^2}{n_2} \right) = \frac{1}{11} \left(22599 - \frac{(503)^2}{12} \right) \\ &= \frac{1}{11} \left(22599 - \frac{253009}{12} \right) = \frac{1}{11} (22599 - 21084.083) = \frac{1}{11} \times 1514.917 = 137.72 \end{aligned}$$

Step 4: Determine the F Statistic

Since $S_2^2 > S_1^2$, we place S_2^2 in the numerator:

$$F = \frac{S_2^2}{S_1^2} = \frac{137.72}{63.66} = 2.164$$

Degrees of freedom: $df_1 = n_2 - 1 = 11$, $df_2 = n_1 - 1 = 9$.

Step 5: Find the Critical Value

For a two-tailed test at $\alpha = 0.05$, we compare F with $F_{0.025}(11, 9)$. From the F-distribution table:

$$F_{0.025}(11, 9) \approx 3.96$$

Step 6: Decision and Conclusion

Since $F = 2.164 < 3.96$, we do not reject H_0 .

Conclusion: There is insufficient evidence to conclude that the population variances are different at the 5% significance level.

14 Example 2: Two Additional F-Test Problems

14.1 Part (a)

Given: $n_1 = 16$, $n_2 = 16$, $s_1^2 = 3.125$, $s_2^2 = 2.400$, $\alpha = 0.05$. Test $H_0 : \sigma_1^2 = \sigma_2^2$ against $H_1 : \sigma_1^2 \neq \sigma_2^2$.

Solution

Since $s_1^2 > s_2^2$, we place s_1^2 in the numerator:

$$F = \frac{s_1^2}{s_2^2} = \frac{3.125}{2.400} = 1.302$$

Degrees of freedom: $df_1 = n_1 - 1 = 15$, $df_2 = n_2 - 1 = 15$.

For a two-tailed test at $\alpha = 0.05$, the critical value is $F_{0.025}(15, 15) \approx 2.86$ (from F-table).

Since $F = 1.302 < 2.86$, we do not reject H_0 . The variances are not significantly different.

14.2 Part (b)

Given: $n_1 = 13$, $n_2 = 41$, $s_1^2 = 6.3$, $s_2^2 = 15.6$, $\alpha = 0.01$. Test $H_0 : \sigma_1^2 = \sigma_2^2$ against $H_1 : \sigma_1^2 < \sigma_2^2$.

Solution

This is a one-tailed (lower-tailed) test. We are testing if σ_1^2 is less than σ_2^2 . However, note that $s_2^2 > s_1^2$, so we set up the F statistic with the larger variance in the numerator:

$$F = \frac{s_2^2}{s_1^2} = \frac{15.6}{6.3} = 2.476$$

Degrees of freedom: $df_1 = n_2 - 1 = 40$, $df_2 = n_1 - 1 = 12$.

For a one-tailed test at $\alpha = 0.01$, we compare F with $F_{0.01}(40, 12)$. From the F-table, $F_{0.01}(40, 12) \approx 3.62$.

Since $F = 2.476 < 3.62$, we do not reject H_0 . There is insufficient evidence to conclude that $\sigma_1^2 < \sigma_2^2$ at the 1% significance level.

Alternatively, note that the alternative hypothesis is $\sigma_1^2 < \sigma_2^2$, which is equivalent to $\frac{\sigma_1^2}{\sigma_2^2} < 1$. We could have computed $F = \frac{s_1^2}{s_2^2} = 0.4038$ and compared with the lower critical value. However, by convention we always take the ratio with the larger variance in the numerator and then compare with the upper critical value. In this case, we are effectively testing $H_1 : \sigma_1^2 < \sigma_2^2$ by checking if $F = \frac{s_2^2}{s_1^2}$ is large enough. Since the computed F is not greater than the critical value, we do not reject H_0 .

15 Summary of Formulas

15.1 Sample Variance

$$S^2 = \frac{1}{n-1} \left(\sum x^2 - \frac{(\sum x)^2}{n} \right)$$

15.2 F-Test Statistic

$$F = \frac{S_1^2}{S_2^2} \quad (\text{with larger variance in numerator})$$

15.3 Degrees of Freedom

- Numerator: $df_1 = n_1 - 1$

- Denominator: $df_2 = n_2 - 1$

15.4 Critical Values

- Two-tailed test: Compare F with $F_{\alpha/2}(df_1, df_2)$
- One-tailed test (upper): Compare F with $F_{\alpha}(df_1, df_2)$

16 Practice Problems

16.1 Problem 1

Two machines produce steel rods. A sample of 10 rods from Machine A has a variance of 2.5 mm^2 , and a sample of 15 rods from Machine B has a variance of 3.2 mm^2 . Test at the 5% significance level whether the variances are equal.

16.2 Problem 2

Given two samples:

Sample 1: 12, 15, 18, 20, 22, 25
Sample 2: 10, 14, 16, 18, 20, 22, 24, 26

Test $H_0 : \sigma_1^2 = \sigma_2^2$ vs $H_1 : \sigma_1^2 > \sigma_2^2$ at $\alpha = 0.05$.

16.3 Problem 3

A researcher wants to compare the variability of two methods. Method A has a sample variance of 8.4 with 10 observations, and Method B has a sample variance of 5.6 with 12 observations. Perform a two-tailed F-test at $\alpha = 0.10$.

16.4 Problem 4

Two brands of batteries are tested for longevity. The variances of lifetime (in hours) are:

Brand X: $n_1 = 8, s_1^2 = 4.2$
Brand Y: $n_2 = 10, s_2^2 = 2.8$

Test whether Brand X has significantly greater variance than Brand Y at $\alpha = 0.05$.

Answer Key (Brief Solutions)

Problem 1:

$F = \frac{3.2}{2.5} = 1.28, df_1 = 14, df_2 = 9, F_{0.025}(14, 9) \approx 3.80$. Since $1.28 < 3.80$, do not reject H_0 .

Problem 2:

Calculate variances: $s_1^2 = 22.0, s_2^2 = 28.0$. Since $s_2^2 > s_1^2$, we test $H_1 : \sigma_1^2 > \sigma_2^2$ by considering $F = \frac{s_2^2}{s_1^2} = 1.273, df_1 = 7, df_2 = 5, F_{0.05}(7, 5) \approx 4.88$. Since $1.273 < 4.88$, do not reject H_0 .

Problem 3:

$F = \frac{8.4}{5.6} = 1.5, df_1 = 9, df_2 = 11, F_{0.05}(9, 11) \approx 2.90$. Since $1.5 < 2.90$, do not reject H_0 .

Problem 4:

We test $H_1 : \sigma_1^2 > \sigma_2^2$. $F = \frac{4.2}{2.8} = 1.5$, $df_1 = 7$, $df_2 = 9$, $F_{0.05}(7, 9) \approx 3.29$. Since $1.5 < 3.29$, do not reject H_0 .

— End of Examples —

16.5 Analogy: Comparing Archer Consistency

- **Equal variances:** Both archers equally consistent
- **Unequal variances:** One archer more erratic than the other
- F -ratio = 1: Equal consistency
- F -ratio $\gg 1$: Significant difference in consistency

17 Practical Applications

17.1 Real-World Examples

Field	Application
Medicine	Testing if new drug reduces blood pressure variability (F-test)
Economics	Analyzing relationship between income and expenditure (regression)
Education	Correlating study time with exam scores (correlation)
Quality Control	Comparing product consistency between two machines (F-test)
Marketing	Predicting sales based on advertising budget (regression)
Psychology	Testing if anxiety levels correlate with sleep quality (correlation)

Table 3: Real-world applications of hypothesis testing

17.2 Common Pitfalls to Avoid

1. **Correlation Causation:** Just because two variables are correlated doesn't mean one causes the other
2. **Outliers:** Extreme values can distort correlation and regression results
3. **Small Sample Size:** May not detect significant relationships
4. **Non-linear Relationships:** These tests only detect linear relationships
5. **Multiple Testing:** Increased chance of Type I error when conducting many tests

18 Summary

18.1 Key Formulas

Test	Formula	When to Use
Regression Slope	$b = \frac{n \sum XY - \sum X \sum Y}{n \sum X^2 - (\sum X)^2}$	Predicting Y from X
Correlation	$r = \frac{n \sum XY - \sum X \sum Y}{\sqrt{[n \sum X^2 - (\sum X)^2][n \sum Y^2 - (\sum Y)^2]}}$	Measuring relationship strength
t-test for r	$t = \frac{r\sqrt{n-2}}{\sqrt{1-r^2}}$	Testing if correlation is significant
F-test	$F = \frac{S_1^2}{S_2^2}$ (larger in numerator)	Comparing two variances

Table 4: Key formulas for hypothesis testing

18.2 Decision Rules

- **Regression:** Reject H_0 if $|t| > t_{\alpha/2, n-2}$
- **Correlation:** Reject H_0 if $|t| > t_{\alpha/2, n-2}$
- **F-test:** Reject H_0 if $F > F_{\alpha}(df_1, df_2)$

18.3 Interpretation Guidelines

r-value	Strength	Interpretation
0.9 to 1.0	Very strong	Highly predictable relationship
0.7 to 0.9	Strong	Clear relationship
0.5 to 0.7	Moderate	Noticeable relationship
0.3 to 0.5	Weak	Slight relationship
0.0 to 0.3	Very weak	Negligible relationship

Table 5: Interpreting correlation coefficients

19 Exercise Problems

19.1 Problem 1: Regression Analysis

The following data shows advertising expenditure (X in \$1000s) and sales (Y in \$1000s):

X	1	2	3	4	5
Y	3	7	8	10	12

Calculate:

- The regression equation $\hat{Y} = a + bX$
- The standard error of estimate
- Test if the regression coefficient is significant at $\alpha = 0.05$

19.2 Problem 2: Correlation Analysis

A study examines the relationship between car age (X in years) and maintenance cost (Y in \$100s):

Age (X)	1	2	3	4	5	6	7	8
Cost (Y)	2	4	5	7	9	10	13	15

Calculate:

- (a) The correlation coefficient
- (b) Test if the correlation is significant at $\alpha = 0.01$
- (c) Interpret the result in practical terms

19.3 Problem 3: F-Test for Variances

Two teaching methods are tested. Exam scores are:

Method A	Method B
85, 88, 90, 92, 95 ($n_1 = 5$)	70, 75, 80, 85, 90, 95, 100 ($n_2 = 7$)

- (a) Calculate variances for both methods
- (b) Perform F-test at $\alpha = 0.05$
- (c) Which method produces more consistent results?

19.4 Problem 4: Comprehensive Analysis

Given the following summary statistics:

$$n = 15, \quad \sum X = 120, \quad \sum Y = 90, \quad \sum XY = 800, \quad \sum X^2 = 1100, \quad \sum Y^2 = 600$$

- (a) Test $H_0 : \rho = 0$ vs $H_1 : \rho \neq 0$ at $\alpha = 0.05$
- (b) Test $H_0 : \beta = 0$ vs $H_1 : \beta \neq 0$ at $\alpha = 0.05$
- (c) Compare the results of (a) and (b) - what do you notice?

19.5 Problem 5: Real-World Application

A pharmaceutical company tests two drug formulations. The active ingredient concentrations (mg/mL) are:

Formulation A	Formulation B
9.8, 10.1, 10.0, 9.9, 10.2 ($n_1 = 5$)	10.0, 9.9, 10.1, 10.0, 10.2, 9.8, 10.1 ($n_2 = 7$)

- (a) Which formulation appears more consistent?
- (b) Perform an F-test to confirm if the difference in consistency is statistically significant at $\alpha = 0.05$
- (c) What recommendations would you make to the company?

Answer Key (Brief Solutions)

Problem 1:

- (a) $b = 2.2, a = 1.0, \hat{Y} = 1.0 + 2.2X$
- (b) $S_{yx} = 0.748$
- (c) $t = 8.385 > t_{0.025,3} = 3.182$, reject H_0

Problem 2:

- (a) $r = 0.992$
- (b) $t = 19.84 > t_{0.005,6} = 3.707$, reject H_0
- (c) Strong positive correlation: older cars require more maintenance

Problem 3:

- (a) $S_A^2 = 14.5, S_B^2 = 116.67$
- (b) $F = 8.046 > F_{0.05}(6, 4) = 6.16$, reject H_0
- (c) Method A is more consistent (lower variance)

Problem 4:

- (a) $r = 0.845, t = 5.72 > 2.160$, reject H_0
- (b) $b = 0.5, t = 5.72 > 2.160$, reject H_0
- (c) Same t-value: testing $\beta = 0$ is equivalent to testing $\rho = 0$ in simple linear regression

Problem 5:

- (a) Formulation A appears more consistent (smaller range)
- (b) $F = 1.24 < F_{0.05}(6, 4) = 6.16$, do not reject H_0
- (c) Both formulations have statistically similar consistency; choose based on other factors like cost

— End of Lecture Notes —